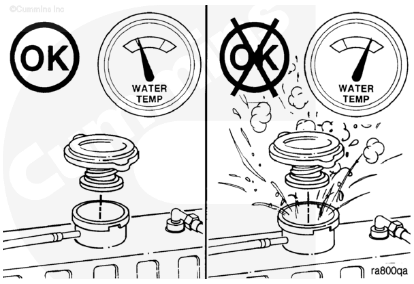
Trainings

Drain

Cooling System - Drain

⚠ WARNING ⚠

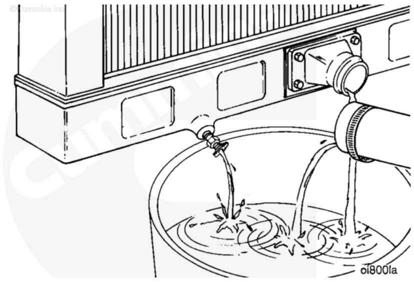
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Remove the radiator cap after the engine is cool.

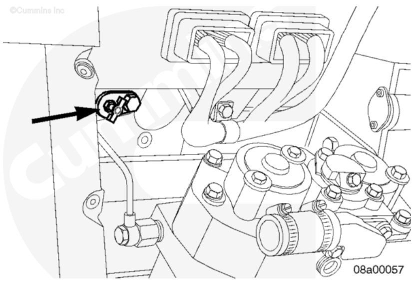
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



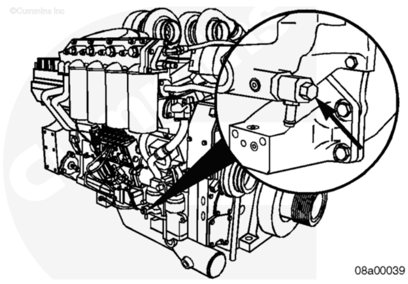
Position the vehicle on level ground. Open the draincock at the bottom of the radiator. Remove the lower radiator hose. Drain the cooling system.

Drain coolant from each bank, if possible.



Left bank: Open draincock on front left bank for applications without an air compressor. Open draincock on air compressor for applications with an air compressor.

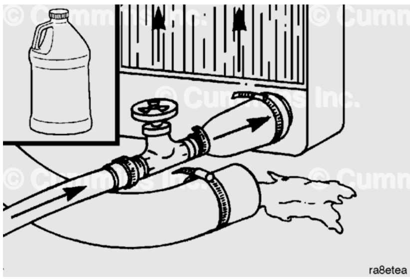
Right bank: Open draincock on water pump, if applicable.



Flush

⚠ CAUTION ⚠

Do not use caustic cleaners in the cooling system. Aluminum components will be damaged.



Every 6000 hours or 2 years of operation, whichever comes first, change the coolant and antifreeze.

The cooling system **must** be clean to work correctly and to eliminate buildup of harmful chemicals.

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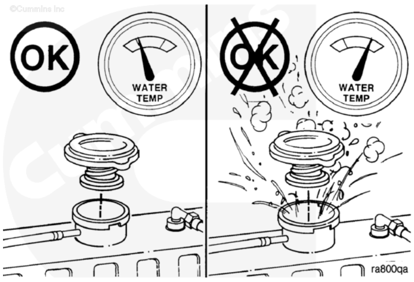
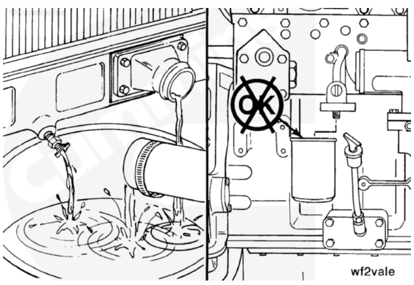


Fleetguard® Restore™ is a heavy-duty cooling system cleaner that removes corrosion products, silica gel, and other deposits. The performance of Restore™ is dependent on time, temperature, and concentration levels. An extremely scaled or flow restricted system, for example, can require higher concentrations of cleaners, higher temperatures, or longer cleaning times or the use of Restore Plus™. Up to twice the recommended concentration levels of Restore™ can be used safely. Restore Plus™ **must** be used **only** at its recommended concentration level. Extremely scaled or fouled systems can require more than one cleaning.

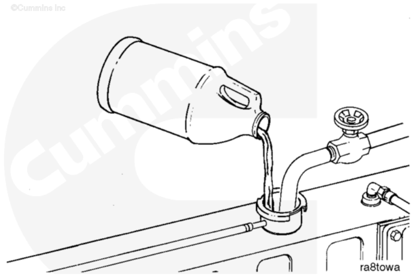
Drain the cooling system. Do **not** allow the cooling system to dry out.

Do **not** remove the coolant filter.



⚠ CAUTION ⚠

Fleetguard® Restore™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation. Engine or radiator damage can occur.

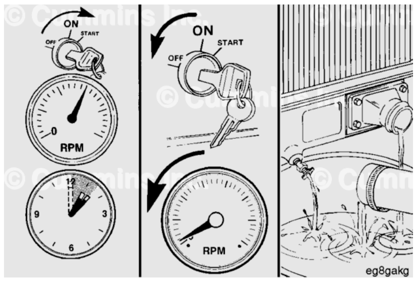


Immediately add 3.8 liters [1 gal] of Fleetguard® Restore™, Restore Plus™, or equivalent, for each 38 to 57 liters [10 to 15 gal] of cooling system capacity, and fill the system with plain water.

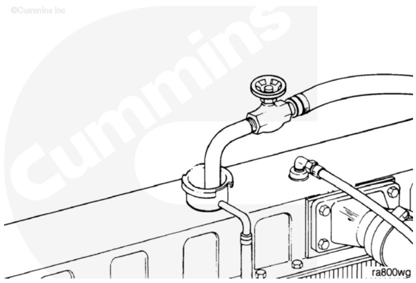
Turn the heater temperature switch to high to allow maximum coolant flow through the heater core. The blower does **not** have to be on.

Operate the engine at normal operating temperatures, at least 85°C [185°F], for 1 to 1-1/2 hours.

Shut off the engine and drain the cooling system.



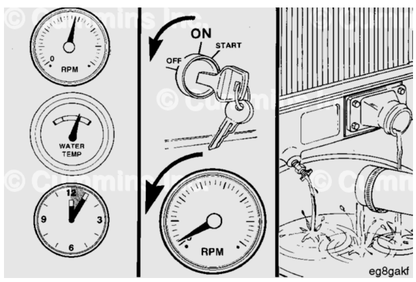
Fill the cooling system with clean water.



Operate the engine at high idle for five minutes with the coolant temperature above 85°C [185°F].

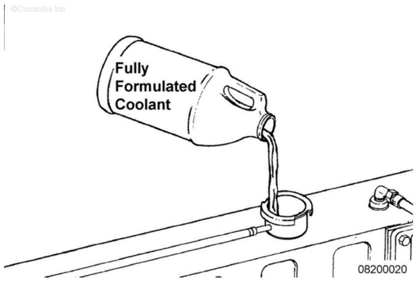
Shut off the engine and drain the cooling system.

If the water being drained is still dirty, the system **must** be flushed again until the water is clean.



Fill the cooling system with **fully-formulated coolant** or a 50/50 mixture of **fully-formulated antifreeze** and good-quality water. Use a service filter to bring the coolant to the correct

SCA concentration level. Refer to 018-018 (/qs3/pubsys2/xml/en/procedures/57/57-018-018.html) (Coolant Specifications) in Section V.



Install the pressure cap.

Operate the engine until it reaches a temperature of 80°C [176°F], and check for coolant leaks.

Leak Test

⚠ WARNING ⚠

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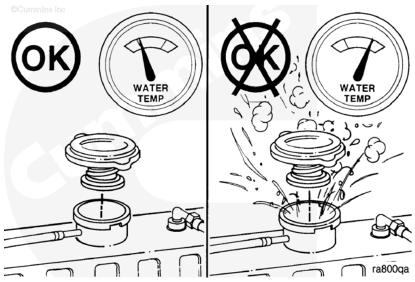
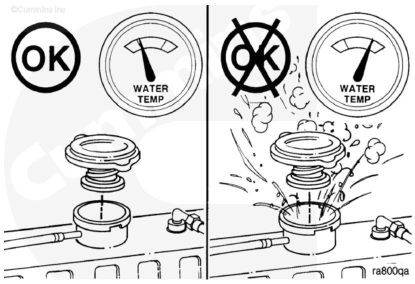


Note : To confirm and locate external coolant leaks, it is not necessary to warm the engine.

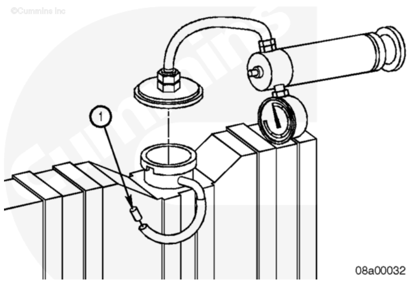
Inspect the coolant for the presence of large amounts of oil. If found, drain the coolant. Remove the oil cooler covers, and inspect for the presence of coolant. If coolant is found, test and replace the oil cooler element.



If no oil contamination is detected, check coolant level and fill, as necessary.

⚠ CAUTION ⚠

Do not apply more than 140 kPa [20 psi] of air pressure to the cooling system. Excessive air pressure can damage the water pump seal.



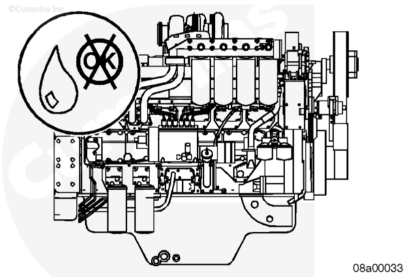
If the radiator is equipped with a pressure relief valve, install a plug in the overflow tube (1).

Install the pressure tester on the radiator fill neck or surge tank (if equipped). Apply maximum air pressure.

Measurements		
	kpa	psi
Maximum Air Pressure:	140	20

Maintain air pressure for at least 5 minutes.

External

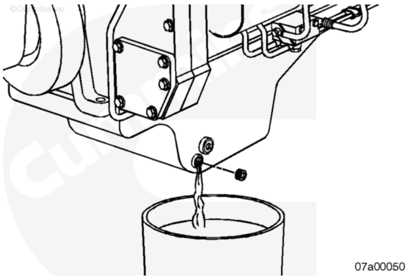


Check for visible coolant leaks at the water pump, air compressor head gasket, oil coolers, aftercoolers, and cylinder heads.

If a leak is detected, repair or replace the leaking component.

If the pressure does **not** change and if no visible external leaks are detected and an internal leak is **not** suspected, remove the test equipment, and replace the radiator cap.

Internal



⚠ WARNING ⚠

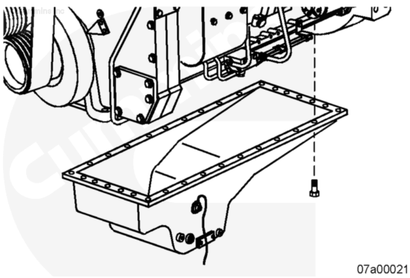
Some state and federal agencies have determined that used engine oil can be carcinogenic and can cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil.

Drain the oil.

Refer to Procedure 007-025 (/qs3/pubsys2/xml/en/procedures/57/57-007-025-tr.html).

Check the oil for the presence of coolant. If the oil does **not** contain more than a trace of coolant, check the aftercoolers.

If the oil contains more than a trace of coolant, remove the oil pan.

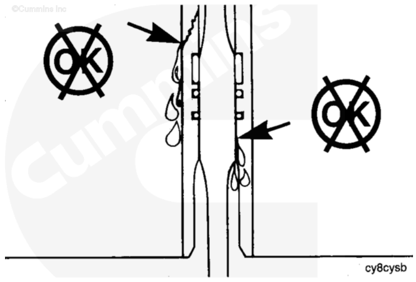


Check for leakage on the inside and outside of the cylinder liner.

Replace all cylinder liners that are leaking on the inside of the liner. Replace the cylinder liner seals if the leak is on the outside of the liner.

Refer to Procedure 001-028 (/qs3/pubsys2/xml/en/procedures/57/57-001-028-tr.html).

If the cylinder liners are **not** leaking, check the aftercooler element.



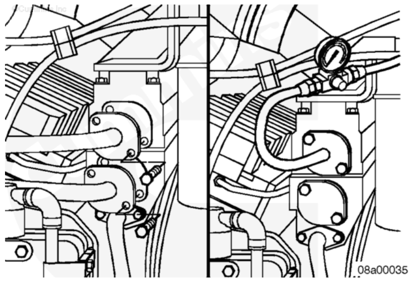
Disconnect the aftercooler coolant hoses. Plug one hose, and apply air pressure to the other hose.

Measurements		
	kpa	psi
Air Pressure:	392	57

If the air pressure decreases, replace the aftercooler element.

Refer to Procedure 010-008 (/qs3/pubsys2/xml/en/procedures/57/57-010-008-tr.html).

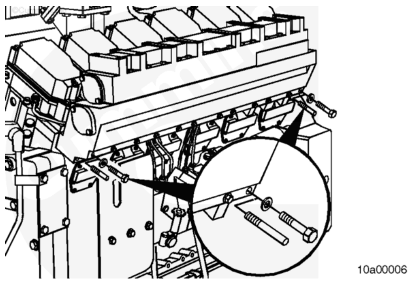
If the air pressure remains the same, check the cylinder heads.



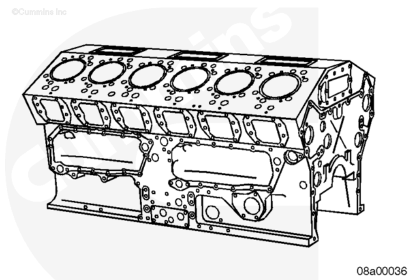
Remove the aftercooler housing. Refer to Procedure 010-002 (/qs3/pubsys2/xml/en/procedures/57/57-010-002-tr.html).

Check the intake ports for coolant.

If coolant is found, replace the cylinder head.



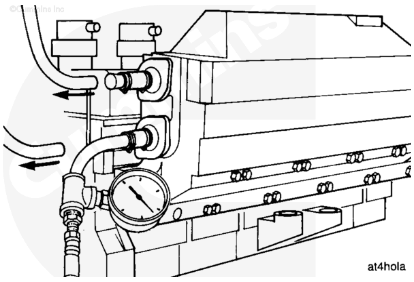
If no leaks are detected, refer to a Cummins Authorized Repair Facility for further testing of the cylinder block.



Remove the test equipment.

Remove the plug from the overflow tube.

Replace the pressure cap.



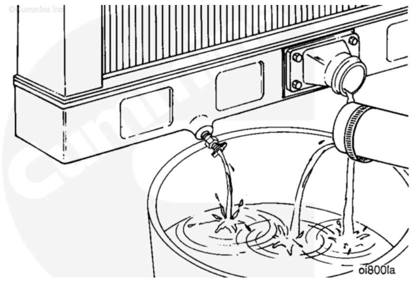
Fill

Close the radiator draincocks.

Install the lower radiator hose(s).

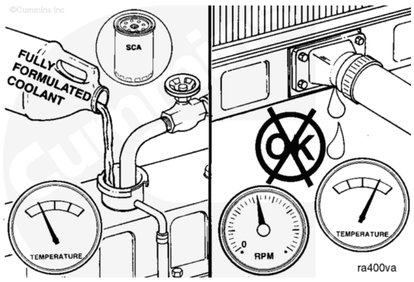
Tighten the hose clamps.

Torque Value: 5 n•m [44 in-lb]

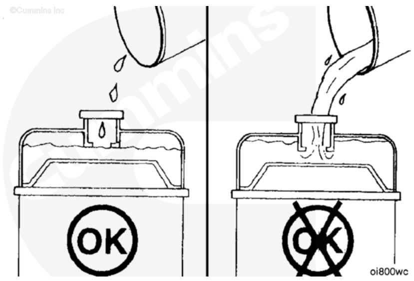


Use fully-formulated coolant to fill the cooling system.

Note : Use the correct units of SCA to obtain the correct cooling system protection.

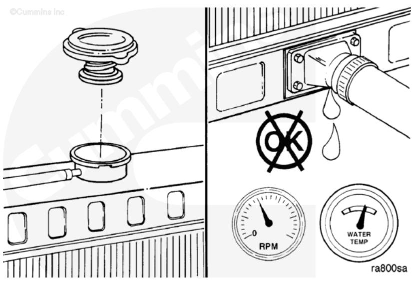


Fill the cooling system with coolant to the bottom of the fill neck in the radiator fill (or expansion) tank.



Replace the radiator fill cap.

Operate the engine until the coolant reaches a temperature of 70°C [158°F]. Check for leaks.



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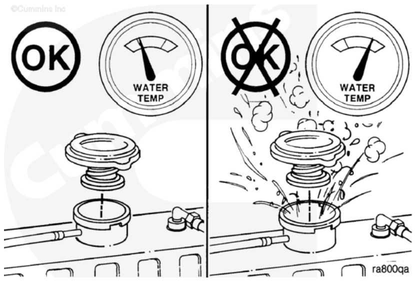

Shut the engine off, and allow it to cool.

Check the coolant level.

Last Modified: 16-Nov-2005